Examen final de Xarxes de Computadors (XC)	Grau en Ingeniería Informàtica	10/4/2026	Primavera 2026

Duration: 1h30m. The quiz will be collected after 25 minutes. The questions have a single correct answer. They are scored p, $-p/3$ if wrong, 0 for no answer.

QUIZ (4 points)

A 2000-byte packet is transmitted at 10 Mbps and the client-server propagation time is 0.5 ms. What will be the minimum delay if there are 3 routers along the way?

1. A. 5.3 ms B. 1.1 ms C. 6.9 ms D. 1.3 ms

Mark the correct answers about IP networks.

2. It is a private network: A. 147.83.50.0/24 B. 150.203.0.0/16 C. 1.0.0.0/8 D. 192.168.50.0/24
3. It is class B: A. 192.168.16.0/24 B. 10.0.0.0/8 C. 172.20.0.0/16 D. 200.0.0.0/24
4. It is a valid subnetwork. A. 180.64.159.98/26 B. 172.16.0.1/16 C. 200.2.80.0/18 D. 172.16.32.0/20

Mark the correct answers about the subnet 10.0.0.0/25.

5. The broadcast address is: A. 10.0.0.127 B. 10.0.0.255 C. 10.255.255.255 D. 10.0.0.128
6. The subnet netmask is: A. 255.255.255.128 B. 255.255.255.127 C. 255.255.255.0 D. 255.0.0.0
7. The largest unicast address is: A. 10.0.0.127 B. 10.255.255.127 C. 10.0.0.128 D. 10.0.0.126

Which is the prefix that aggregates the following networks with the largest mask?

8. 172.16.64.0/20, 172.16.80.0/21: A. 172.16.64.0/19 B. 172.16.32.0/18 C. 172.16.0.0/16 D. 172.16.64.0/18
9. 192.168.32.0/20, 192.168.48.0/21, 192.168.40.0/22: A. 192.168.32.0/18 B. 192.168.0.0/18 C. 192.168.48.0/19 D. 192.168.32.0/19

Mark the correct answers about the following protocols.

10. IP: A. It is a layer 4 protocol of the OSI model. B. It guarantees that all packages will be received by the destination. C. If there is fragmentation, fragments of the same packet may have different TTLs. D. The header has an error detection field for errors that occur to the entire IP packet.
11. ICMP: A. Allows to detect package duplication conflicts. B. ICMP messages are encapsulated in IP datagrams. C. ICMP packets do not pass through a router with PNAT. D. ICMP messages are encapsulated in layer 4 segments.
12. ARP: A. The ARP Reply message is broadcast. B. It allows to find out the IP address of a network interface given the MAC address. C. The ARP broadcast address is 255.255.255.255. D. More than one device can respond to an ARP Request message, but it indicates duplicated IP.
13. RIP: A. A directly connected network is at 0 hops (metric 0, no GW). B. The infinite metric (Destination unreachable) is 15. C. Each node sends updates to all routers in the network. D. All false

In a network with RIPv2, one of the routers has the following routing table. Mark the correct answer.

#	D	Gw	M	If
1	10.10.0.0/16	-	1	e1
2	10.20.0.0/16	-	1	e2
3	200.50.50.0/24	R1	3	e0
4	0.0.0.0/0	R1	2	e0

14. Directly connected networks: A. (0.0.0.0/0, 10.10.0.0/16) B. (10.10.0.0/16, 10.20.0.0/16) C. (200.50.50.0/24, 10.20.0.0/16) D. (10.10.0.0/16, 200.50.50.0/24)

Split-horizon is activated. Which will be the update messages it will send through the indicated interfaces? Notation: {(D; M), $\cdots$}.

15. e0: A. {(10.0.0.0/8; 2), (200.50.50.0/24; 4), (0.0.0.0/0; 3)} B. {(10.10.0.0/16; 2), (10.20.0.0/16; 2)} C. {(10.10.0.0/16; 1), (10.20.0.0/16; 1)} D. {(10.10.0.0/16; 1), (10.20.0.0/16; 1), (200.50.50.0/24; 3), (0.0.0.0/0; 2)}
16. e1: A. {(10.10.0.0/16; 2), (10.20.0.0/16; 2), (200.50.50.0/24; 4), (0.0.0.0/0; 3)} B. {(10.20.0.0/8; 1), (200.50.50.0/24; 3), (0.0.0.0/0; 2)} C. {(10.20.0.0/8; 2), (200.50.50.0/24; 4), (0.0.0.0/0; 3)} D. {(10.20.0.0/16; 1), (200.50.50.0/24; 3), (0.0.0.0/0; 2)}
17. e2: A. {(10.10.0.0/16; 1), (200.50.50.0/24; 3), (0.0.0.0/0; 2)} B. {(10.0.0.0/8; 1)} C. {(10.10.0.0/8; 1), (200.50.50.0/24; 3), (0.0.0.0/0; 2)} D. {(10.10.0.0/16; 2), (10.20.0.0/16; 2), (200.50.50.0/24; 4), (0.0.0.0/0; 3)}

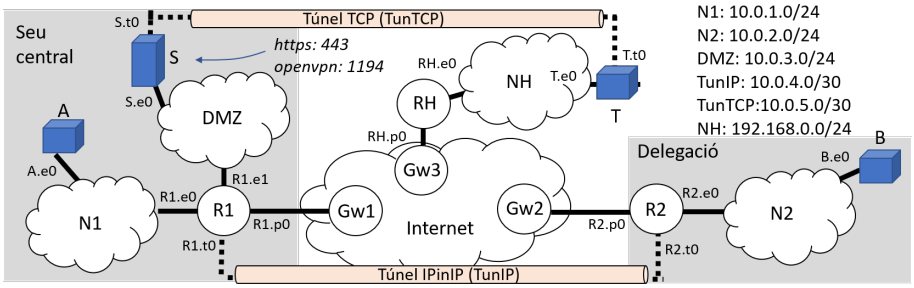
Mark the correct statements about IP security.

18. ACLs: A. An ACL is an ordered list of rules applied to all packets entering or leaving a router interface. B. They allow to inspect the complete contents of the package (payload) to block specific applications. C. They are applied only to filter traffic leaving internal networks towards the Internet. D. They are used to filter datagrams to prevent them from leaving or entering a router based on information that is only found in the IP header.
19. Tunnels: A. Adding a tunnel does not affect the routing table of the device. B. They allow to establish secure connections by encapsulating packets within other protocols to protect the content between two networks. C. They are only used to translate IP addresses within a private network. D. The network interface of a tunnel is a physical interface.
20. PNAT: A. All false B. It changes the source IP address and port on packets that enter from the public network to a private one, while maintaining the destination IP address and port. C. It changes the source IP address and port on packets leaving a private network to the public one, while maintaining the destination IP address and port. D. It changes the destination IP address while maintaining the destination port on packets leaving a private network to the public one.

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Problem IP
(6 points)



The figure shows a corporate network that connects a headquarters and a branch office through an IPinIP tunnel. It also allows remote workers to connect to the corporate network via an IP-over-TCP tunnel (OpenVPN), which terminates at server S. When S extracts the packets form the tunnel, it keeps the address of T in the IP header. Worker T has established a tunnel of this type with S. Router RH also includes a PNAT functionality. Communication between equipment at the headquarters and the branch office always goes through the IPinIP tunnel. All equipment on headquarters and branch office accesses the Internet through router R1 (when branch office equipment accesses the Internet, it must also use the tunnel). Server S is located in a DMZ and runs HTTPS (443) and VPN (1194) accessible from Internet. Names can be resolved (53). We have configured PNAT at R1 and have firewalls for inbound and outbound traffic on interface p0 at both R1 and R2 (for outbound traffic, the PNAT acts first followed by the ACL. Reverse order for inbound traffic).

Which routing table at R1, R2, RH, B, S and T is correct? (Net/mask, Gw, Iface)

1. R1: A. (N1, -, e0), (DMZ, -, e1), (TunIP, -, t0), (N2, Gw1, p0), (0.0.0.0/0, Gw1, p0)
   B. (N1, -, e0), (DMZ, -, e1), (N2, Gw1, p0), (0.0.0.0/0, Gw1, p0)
   C. (N1, -, e0), (DMZ, -, e1), (TunIP, -, t0), (N2, R2.t0, t0), (Gw1/32, -, p0), (TunTCP, S.e0, e1), (0.0.0.0/0, Gw1, p0)
   D. (N1, -, e0), (DMZ, -, e1), (TunIP, -, t0), (0.0.0.0/0, R2.t0, t0)
2. R2: A. (N2, -, e0), (TunIP, -, t0), (0.0.0.0/0, R1.t0, t0)
   B. (N2, -, e0), (TunIP, -, t0), (NH, Gw2, p0), (0.0.0.0/0, R1.t0, t0), (R1.p0/32, Gw2, p0)
   C. (N2, -, e0), (TunIP, -, t0), (TunTCP, Gw2, p0), (Gw2/32, -, p0), (0.0.0.0/0, R1.t0, t0)
   D. (N2, -, e0), (TunIP, -, t0), (Gw2/32, -, p0), (0.0.0.0/0, R1.t0, t0), (R1.p0/32, Gw2, p0)
3. RH: A. (NH, -, e0), (TunTCP, T.e0, e0), (0.0.0.0/0, Gw3, RH.p0)
   B. (NH, -, e0), (DMZ, T.e0, e0), (Gw3/32, -, p0), (0.0.0.0/0, Gw3, RH.p0)
   C. (NH, -, e0), (Gw3/32, -, p0), (0.0.0.0/0, Gw3, RH.p0)
   D. (NH, -, e0), (N2, Gw3, p0), (0.0.0.0/0, Gw3, RH.p0)
4. B: A. (N2, -, e0), (N1, R2.e0, e0), (DMZ, R2.e0, e0)
   B. (N2, -, e0), (NH, R2.e0, e0), (0.0.0.0/0, R2.e0, e0)
   C. (N2, -, e0), (TunTCP, R2.e0, e0), (0.0.0.0/0, R2.e0, e0)
   D. (N2, -, e0), (0.0.0.0/0, R2.e0, e0)
5. S: A. (DMZ, -, e0), (TunTCP, -, t0), (0.0.0.0/0, R1.e1, e0)
   B. (DMZ, -, e0), (NH, -, t0), (0.0.0.0/0, R1.e1, e0)
   C. (DMZ, -, e0), (NH, -, t0), (TunTCP, -, t0), (0.0.0.0/0, R1.e1, e0)
   D. (DMZ, -, e0), (0.0.0.0/0, R1.e1, e0)
6. T: A. (NH, -, e0), (TunTCP, -, t0), (10.0.0.0/8, S.t0, t0), (0.0.0.0/0, RH.e0, e0)
   B. (NH, -, e0), (0.0.0.0/0, RH.e0, e0)
   C. (NH, -, e0), (TunTCP, -, t0), (0.0.0.0/0, RH.e0, e0)
   D. (NH, -, e0), (0.0.0.0/32, S.t0, t0)

We run the ping command. State which combination of headers at the OUTPUT of the corresponding interfaces is correct. Notation: IP: (@src, @dst, Protocol), UDP: (Port_src, Port_dst). In case of two IP headers, the outer header is written first.

7. B> ping A: A. R2.p0: (B.e0, A.e0, ICMP) B. R1.e0: (R2.p0, R1.p0, IPinIP), (B.e0, A.e0, ICMP) C. R2.p0: (R2.p0, R1.p0, ICMP) D. R2.p0: (R2.p0, R1.p0, IPinIP), (B.e0, A.e0, ICMP)
8. T> ping A: A. R1.e0: (T.t0, A.e0, ICMP) B. RH.p0: (T.e0, S.e0, UDP), (30000, 1194), (S.t0, A.e0, ICMP) C. R1.e0: (R1.p0, A.e0, ICMP) D. RH.p0: (RH.p0, A.e0, ICMP)

State which ACL is correct (neglect ICMP). Notation: (Protocol, Net_org, Net_dest, Port_org, Port_dst, Action)

9. R1.p0 OUT: A. (TCP, any, any, >= 1024, any, Accept), (UDP, R1.p0, -, >=1024, 53, Accept), (IP, any, any, any, any, Deny)
   B. (TCP, any, any, >= 1024, any, Accept), (IPinIP, R1.p0, R2.p0, -, -, Accept), (IP, any, any, any, any, Deny)
   C. (TCP, any, any, >= 1024, any, Accept), (UDP, R1.p0, -, >=1024, 53, Accept), (IP, any, any, any, any, Deny)
   D. (TCP,any,any,>= 1024, any,Accept), (IPinIP,R1.p0,R2.p0,-,-,Accept), (UDP, R1.p0, -, >=1024, 53, Accept), (TCP, R1.p0, any, 1194, any, Accept), (TCP, R1.p0, any, 443, any, Accept), (IP, any, any, any, any, Deny)
10. R2.p0 IN: A. (IPinIP, R1.p0, R2.p0, -, -, Accept), (UDP, R2.p0, any, 1194, any, Accept), (TCP, R2.p0, any, 443, any, Accept), (IP, any, any, any, any, Deny)
    B. (IP, any, any, any, any, Deny), (UDP, R2.p0, -, >=1024, 53, Accept), (IPinIP, R1.p0, R2.p0, -, -, Accept)
    C. (IPinIP, R1.p0, R2.p0, -, -, Accept), (IP, any, any, any, any, Deny)
    D. (IP, any, any, any, any, Deny), (IPinIP, R1.p0, R2.p0, -, -, Accept), (TCP, R2.p0, any, 1194, any, Accept), (TCP, R2.p0, any, 443, any, Accept)